

Subject: IND320

Student name: Abdul Haadi Sheikh

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In this first part of the project I worked with both Jupyter Notebook and Streamlit to get a basic dashboard up and running. In Jupyter Notebook I read the supplied CSV file with Pandas, printed the contents in a structured way and created plots of both individual columns and all columns together. I made sure to run all code cells so the results and figures are visible when I export the notebook to PDF. I also added comments so the code is easier to understand and follow if someone else were to continue my work.

For the Streamlit app I created four pages, with a home page, a data table page, a plot page and an about page. The home page has a sidebar that lets you navigate to the other pages. On the data table page the CSV file is shown together with a small line chart for each column for the first month of the dataset. On the plot page it is possible to choose one column or all columns from a drop down menu and also filter months with a slider. The app uses caching so that the CSV file loads faster, and I added the CSV file into the repository so that it works both locally and on Streamlit Cloud.

I struggled a lot with file formats in the beginning. I first worked only with ipynb files but Streamlit does not accept those, so I had to make sure to keep the code in py files as well. I also had problems when I tried to organize everything into folders, because Streamlit Cloud and GitHub did not always find the files in the right place. In the end I fixed this by keeping streamlit_app.py and utils.py in the root folder, and by using relative paths instead of absolute ones for loading the CSV file. I also had to clean up the requirements.txt file so that only the packages I really needed were listed. After these changes the deployment on Streamlit Cloud finally worked.

I used AI tools like ChatGPT while working on the project. It helped me when I got stuck on errors, for example with missing files, wrong imports and when the app failed to run in the cloud. It also explained how to set up caching, how to structure the repository and how to read the Streamlit error logs. I learned more about how to debug and organize a project like this.

My GitHub repository is available here: https://github.com/Sheikh2003/IND320_1_Project

My Streamlit app is available here: <https://streamspinn.streamlit.app>